Department of Mechanical Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. Mechanical Engineering
Course Code & Title: CE3391 Fluid Mechanics and Machinery
Name of the Faculty member (s): Dr.P.Sureshkumar, ASCP/Mechanical Engineering

Innovative Practice Description

- **Unit / Topic:** Unit IV / CE3391 Fluid Mechanics and Machinery
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO18 & TLO19
- **Activity Chosen:** Theory to Practice
- **Justification:**
  - ✓ To understand the classification of turbine based on various parameters
  - ✓ To explain the theory of roto-dynamics machines
  - ✓ To recollect the theoretical concepts of turbine and its working principles by practical experience.
- **Time Allotted for the Activity:** 50 minutes for listening concepts and hands on experience in operating turbine by understanding the different turbine and its working conditions.
- **Details of the Implementation:**

  The students were asked to operate the turbine after listening the concepts of working principle of turbine. Students skills were assessed by interacting with them by asking questions related to concepts turbine.

- **CO – PO / PSO mapping:**

CO	PO5	PO7	PO8	PO9	PO10	PO12	PSO3
CO4	2	2	1	2	1	3	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

CO	PO5	PO9	PO10	PSO3
CO4	To select the appropriate turbine of the specific application for performing complex engineering activities	To carry out the hands on practice on operating the turbine by group of students	To utilize the proper turbine for a particular application in their field for future technological changes	To design and analyze the hydraulic system

- **Images / Screenshot of the practice:**

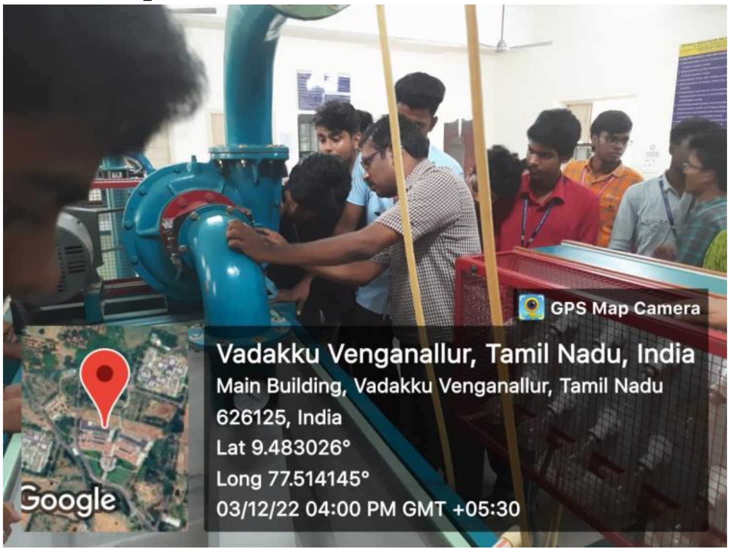

Figure 1: Hands on training on performance of Kaplan turbine

Figure 2: Hands on training on performance of Orifice meter and venturimeter

Figure 3: Hands on training on performance of Francis turbine

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    Concepts were clearly understood by the students. They understand the theoretical concept of turbine by obtaining the hands on experience in operating the turbine at different load conditions.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - ✓ By the hands on training in turbine, students can able to understand the theoretical velocity triangles, blade angle, shapes and the performance of turbine easily.
    - ✓ By interacting with fellow students, they improved the communication skills and team building activities.
- ❖ ***Challenges faced in implementation:***
  Short duration period for the hands on training of operating the turbine is the huge challenge while the students are involving the real time practical session.

References:

1. Modi P.N. and Seth, S.M. "Hydraulics and Fluid Mechanics", Standard Book House, New Delhi 2013.
2. Graebel. W.P, "Engineering Fluid Mechanics", Taylor & Francis, Indian Reprint, 2011
3. Kumar K. L., "Engineering Fluid Mechanics", Eurasia Publishing House(p) Ltd., New Delhi 2016
4. Robert W.Fox, Alan T. McDonald, Philip J.Pritchard, "Fluid Mechanics and Machinery", 2011.
5. Streeter, V. L. and Wylie E. B., "Fluid Mechanics", McGraw Hill Publishing Co. 2010.
6. Dr.R.K.Bansal "Fluid Mechanics and Hydraulic Machines, Laxmi Publications (P) Ltd., 2017.
7. Frank M.White, "Fluid Mechanics" Tata McGraw Hill Education Private Limited, 2012.

CE3391 Fluid Mechanics and Machinery - Innovative Practice - Theory to Practice

39 responses

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Name of the student

39 responses

Kiruthick N

Prem Sukumar.M

M.subbiah@selvam

KRISHNA.R

Gowtham J

SRI SABARI VENKATESH R

C.palpandi

JEYA PRAKASH K

Balaji S

Harish krishnan R

A. Kodiappan

M.Surya Prakash

KRISHNA R

Krishnakumar

S. Kadar mohaideen

SIVAPRAKASH P

Vignesh.B

K.PAVAN SHANKAR

Muthukumar R

Sibi Siddharth

SARAVANA KUMAR S

Sudarshan R

Balaji A

S.surya

Varun K

Ramji N

N.Gurusankar

HARISHH V

M.Bhuvaneswaran

M. Aathithyan

P.Ganesh Kumar

Sibirakesh S

R.K.RAMNATH SARGURU

Ramji.R

S sriram

S.Sankarganesh

S.JAWAHAR GANESAN

M.Santhosh kumar

V.Sateesh Bharathi

Register No.

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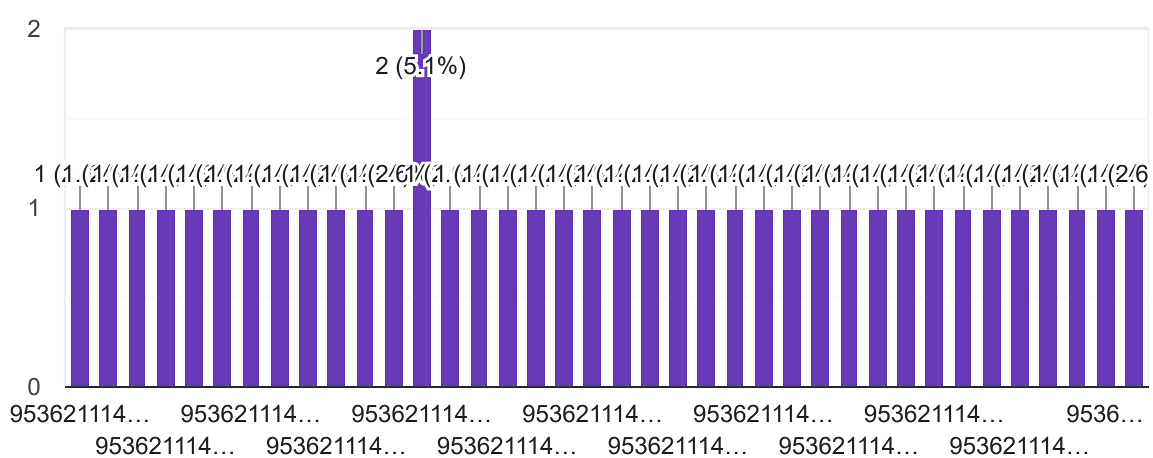

1. Instructor clarifies diffult aspects of this innovative activity (1- low, 5- high)

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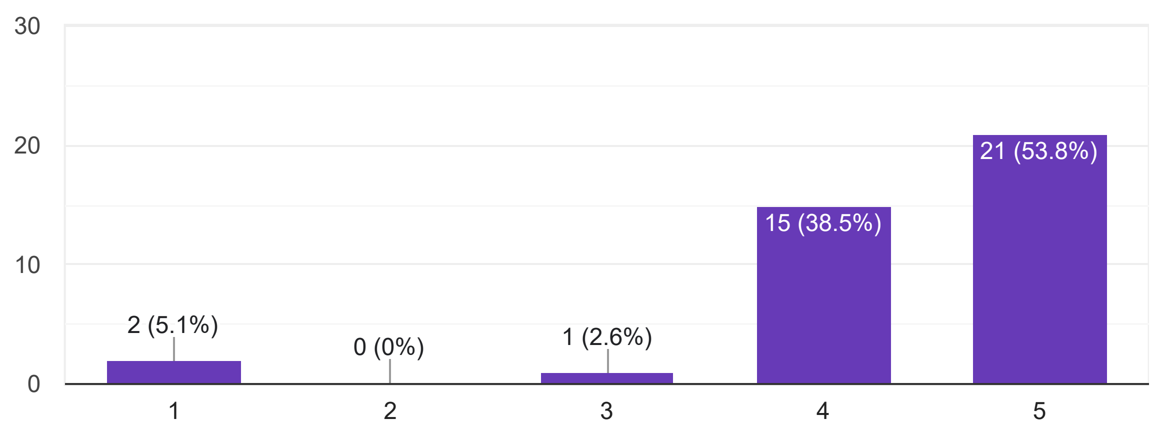

2. This innovative activity improves my opinion about the content of the subject (1- low, 5-high)

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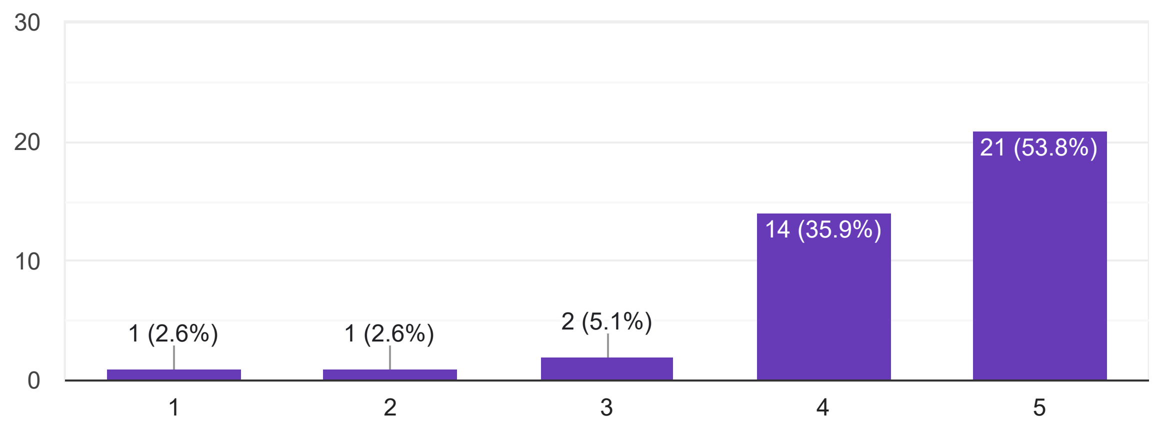

3. I find new information about the topics and subjects sing new technologies (1- low, 5-high)

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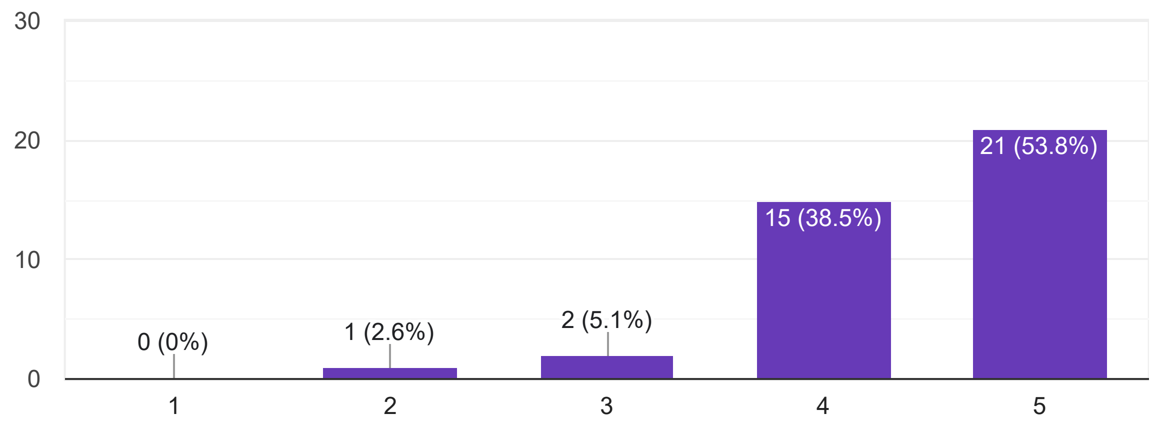

4. I suggest this innovative practice to teach the topic for forthcoming students (1- low, 5-high)

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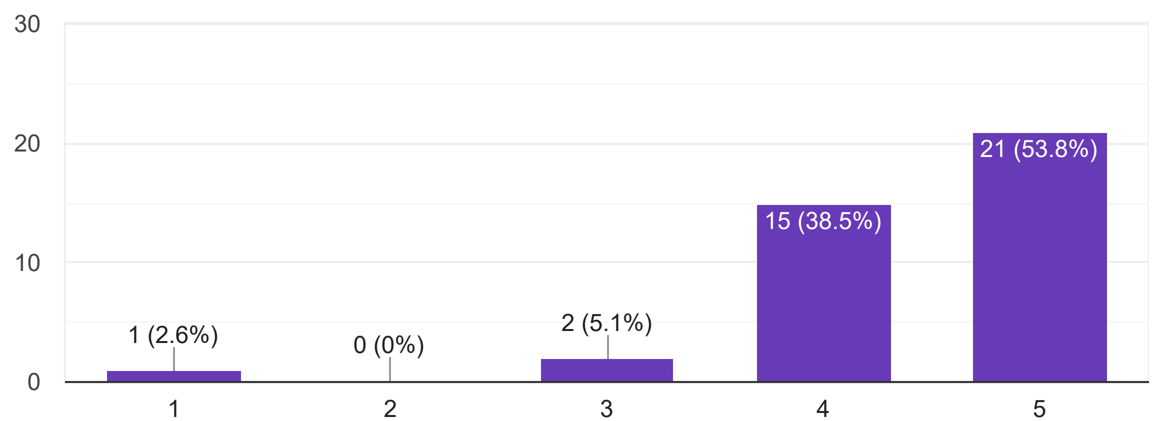

5. This innovative activity builds any self confidence to understand the content of the delivery (1- low, 5-high)

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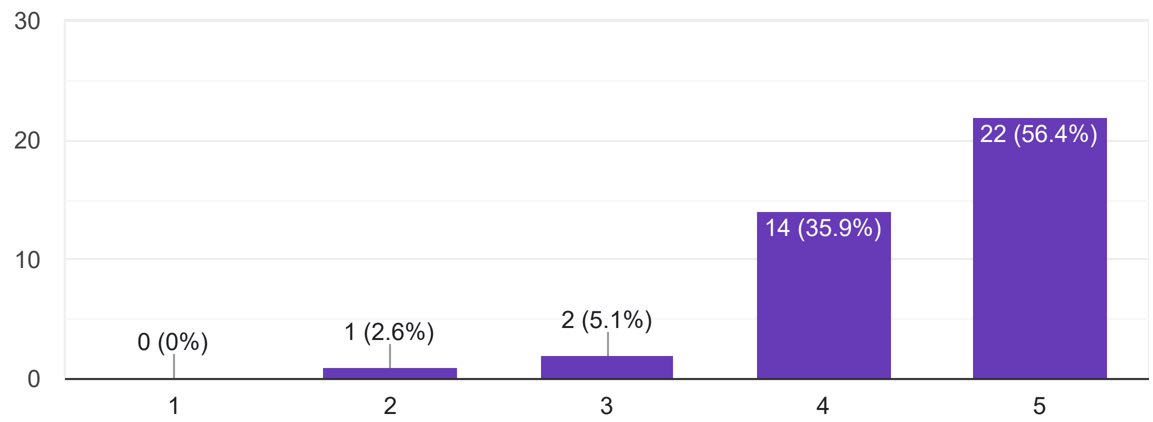

The most useful thing/skill i learned from this activity was

39 responses

All of us

All

I learned basic about turbine

Calculation

Which is very useful to seen by virtual

Easily learning the concept

Self confidence

Useful informations about turbines and its working models

Yes

Real time experience and exploring about turbines and other measurement devices.

To see the instruments on real

Hands-on practice and learning about draft tube

Visualize the turbines, and gain a some Theoretical knowledge.

Self management

Easily learning concept

It’s is useful

Learned new things about turbine

Yes

Turbine types in lab

.

Improving From this...

Turbine

Kaplan turbine

Kaplan turbine

Gained some skills

Pelton wheel

Fluid and its properties

Lab Session is very useful to me

Got real time experience

Learning skill useful

Calculation

Machinery

Turbine

Turbines

Skill

Phelton wheel

discipline